

WeBWorK assignment number WW_04 is due : 08/04/2008 at 11:00pm CDT.

The

(* replace with url for the course home page *)

for the course contains the syllabus, grading policy and other information.

This file is /conf/snippets/setHeader.pg you can use it as a model for creating files which introduce each problem set.

The primary purpose of WeBWorK is to let you know that you are getting the correct answer or to alert you if you are making some kind of mistake. Usually you can attempt a problem as many times as you want before the due date. However, if you are having trouble figuring out your error, you should consult the book, or ask a fellow student, one of the TA's or your professor for help. Don't spend a lot of time guessing – it's not very efficient or effective.

Give 4 or 5 significant digits for (floating point) numerical answers. For most problems when entering numerical answers, you can if you wish enter elementary expressions such as $2 \wedge 3$ instead of 8, $\sin(3 * \pi/2)$ instead of -1, $e \wedge (\ln(2))$ instead of 2, $(2 + \tan(3)) * (4 - \sin(5)) \wedge 6 - 7/8$ instead of 27620.3413, etc. Here's the [list of the functions](#) which WeBWorK understands.

You can use the Feedback button on each problem page to send e-mail to the professors.

1. (1 pt) 2360lib/chap4/leon.4.3/n01.pg

(S. 4-3, cf. no. 1)

For each of the following linear transformations $L : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, determine the matrix A representing L with respect to the standard basis $\{\mathbf{e}_1, \mathbf{e}_2\}$ and the matrix B representing L with respect to $[\mathbf{u}_1 = (1, 1)^T, \mathbf{u}_2 = (1, 2)^T]$, and a matrix S such that $B = S^{-1}AS$.

(a) $L(\mathbf{x}) = (x_1, x_1 + x_2)^T$

$A =$ _____

$B =$ _____

$S =$ _____

(b) $L(\mathbf{x}) = (x_2, x_1)^T$

$A =$ _____

$B =$ _____

$S =$ _____

2. (1 pt) 2360lib/chap4/leon.4.3/n05.pg

(S. 4-3, cf. no. 5)

Let $L : P_3 \rightarrow P_3$ be defined by

$$L(p(x)) = p(0)x^2 + p'(0)x + p''(0)$$

(a) Find the matrix A representing L with respect to the standard basis $[x^2, x, 1]$.

$A =$ _____

(b) Find the transition matrix S representing the change of coordinates from the basis $[x^2, x + 1, x - 1]$ to the standard basis $[x^2, x, 1]$.

$S =$ _____

(c) Use this to find the matrix B representing L with respect to the basis $[x^2, x + 1, x - 1]$ by $B = S^{-1}AS$.

$B =$ _____

3. (1 pt) 2360lib/chap5/leon.5.1/n02.pg

(S. 5-1 cf. no. 2)

Let $\mathbf{v} = (2, 1, 3)^T$ and $\mathbf{w} = (6, 3, 9)^T$

The scalar projection of $\mathbf{v}$ onto $\mathbf{w}$ is:

and the vector projection of $\mathbf{v}$ onto $\mathbf{w}$ is:

Let $\mathbf{v} = (-2, 3, 1)^T$ and $\mathbf{w} = (1, 2, 4)^T$

The scalar projection of $\mathbf{v}$ onto $\mathbf{w}$ is:

and the vector projection of $\mathbf{v}$ onto $\mathbf{w}$ is:

4. (1 pt) 2360lib/chap5/leon.5.1/n08.pg

(S. 5-1 cf. no. 8)

Find the equation of the plane normal to the given vector $\mathbf{N}$ and passing through the point P_0 . (Use x, y, z as variables.)

$\mathbf{N} = (3, 1, 2), \quad P_0 = (1, 1, 1)$

equation: _____ = 0

5. (1 pt) 2360lib/chap5/leon.5.1/n09.pg

(S. 5-1 cf. no. 9)

Find the distance from the point $(1, 2, 3)$ to the plane $3x + 2y + z = 0$.

6. (1 pt) 2360lib/chap5/leon.5.1/n10.pg
(S. 5-1 cf. no. 10)

Find the distance from the point $(1, 2, 3)$ to the plane $(x - 2) + 2(y - 1) - 2(z + 1) = 0$.

7. (1 pt) 2360lib/chap5/leon.5.1/ur.la.17.1.pg

Let $x = \begin{bmatrix} 3 \\ -3 \\ -5 \end{bmatrix}$ and $y = \begin{bmatrix} 3 \\ 5 \\ 2 \end{bmatrix}$.

Find the dot product of x and y .

$x \cdot y =$ _____.

8. (1 pt) 2360lib/chap5/leon.5.1/ur.la.17.2.pg

Let $x = \begin{bmatrix} -1 \\ 3 \\ 5 \\ -3 \end{bmatrix}$.

Find the norm of x and the unit vector in the direction of x .

$\|x\| =$ _____,
 $u = \begin{bmatrix} \text{_____} \\ \text{_____} \\ \text{_____} \\ \text{_____} \end{bmatrix}$.

9. (1 pt) 2360lib/chap5/leon.5.1/ur.la.17.3.pg

Find the angle α between the vectors $\begin{bmatrix} 3 \\ 2 \\ 3 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}$.

$\alpha =$ _____.

10. (1 pt) 2360lib/chap5/leon.5.1/ur.la.17.6.pg

Find a vector v perpendicular to the vector $u = \begin{bmatrix} -2 \\ -9 \end{bmatrix}$.

$v = \begin{bmatrix} \text{---} \\ \text{---} \end{bmatrix}$.

11. (1 pt) 2360lib/chap5/leon.5.1/ur.la.17.7.pg

Find the value of k for which the vectors

$x = \begin{bmatrix} -4 \\ -2 \\ -2 \\ -5 \end{bmatrix}$ and $y = \begin{bmatrix} 1 \\ 2 \\ 0 \\ k \end{bmatrix}$ are orthogonal.

$k =$ _____.

12. (1 pt) 2360lib/chap5/leon.5.1/ur.la.17.10.pg

Find the length of the vector $x = \begin{bmatrix} -6 \\ 5 \\ 3 \end{bmatrix}$.

$\|x\| =$ _____.

13. (1 pt) 2360lib/chap5/leon.5.2/n01.pg
(S. 5-2 cf. no. 1)

Find a basis for each of the indicated subspaces. In case the subspace is trivial (consisting only of the zero vector) enter NONE . Use the angular bracket notation ($\langle$ and $\rangle$) for vectors.

(a)

$$A = \begin{bmatrix} 1 & 4 \\ 3 & 12 \end{bmatrix}$$

a basis for $R(A^T)$: _____

a basis for $N(A)$: _____

a basis for $R(A)$: _____

a basis for $N(A^T)$: _____

(b)

$$A = \begin{bmatrix} 1 & 1 & 4 \\ 2 & 0 & 5 \end{bmatrix}$$

a basis for $R(A^T)$: _____

a basis for $N(A)$: _____

a basis for $R(A)$: _____

a basis for $N(A^T)$: _____

14. (1 pt) 2360lib/chap5/leon.5.2/n02.pg
(S. 5-2 cf. no. 2)

In R^3 let $S = \text{span}((1, 2, 4)^T)$.

A basis for $S^\perp$ is: _____

15. (1 pt) 2360lib/chap5/leon.5.2/n04.pg
(S. 5-2 cf. no. 4)

In R^4 let $S = \text{span}((1, 0, 1, 1)^T, (0, 2, 1, 0)^T)$.

A basis for $S^\perp$ is: _____

16. (1 pt) 2360lib/chap5/leon.5.2/ur.la.17.14.pg

Find two linearly independent vectors perpendicular to the vec-

tor $v = \begin{bmatrix} -8 \\ 3 \\ 1 \end{bmatrix}$.

$\begin{bmatrix} \text{---} \\ \text{---} \\ \text{---} \end{bmatrix}, \begin{bmatrix} \text{---} \\ \text{---} \\ \text{---} \end{bmatrix}$.

17. (1 pt) 2360lib/chap5/leon.5.2/ur_la.17.15.pg

Find the orthogonal projection of $v = \begin{bmatrix} 18 \\ -5 \\ -15 \end{bmatrix}$ onto the subspace V of $\mathbb{R}^3$ spanned by $\begin{bmatrix} 2 \\ 4 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} -6 \\ -1 \\ 8 \end{bmatrix}$.

$\text{proj}_V(v) = \begin{bmatrix} \rule{1cm}{0.4pt} \\ \rule{1cm}{0.4pt} \\ \rule{1cm}{0.4pt} \end{bmatrix}$.

18. (1 pt) 2360lib/chap5/leon.5.2/ur_la.17.16.pg

Find the orthogonal projection of $v = \begin{bmatrix} 5 \\ 14 \\ -14 \\ -19 \end{bmatrix}$ onto the subspace V of $\mathbb{R}^4$ spanned by $\begin{bmatrix} 4 \\ -4 \\ -1 \\ 3 \end{bmatrix}$ and $\begin{bmatrix} 4 \\ 1 \\ 12 \\ 0 \end{bmatrix}$.

$\text{proj}_V(v) = \begin{bmatrix} \rule{1cm}{0.4pt} \\ \rule{1cm}{0.4pt} \\ \rule{1cm}{0.4pt} \\ \rule{1cm}{0.4pt} \end{bmatrix}$.

19. (1 pt) 2360lib/chap5/leon.5.2/ur_la.17.18.pg

Among all the unit vectors $u = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ in $\mathbb{R}^3$, find the one for which the sum $x + 8y + 4z$ is minimal.

$u = \begin{bmatrix} \rule{1cm}{0.4pt} \\ \rule{1cm}{0.4pt} \\ \rule{1cm}{0.4pt} \end{bmatrix}$.

20. (1 pt) 2360lib/chap5/leon.5.2/ur_la.18.1.pg

Find the missing coordinates such that the three vectors form an orthonormal basis for $\mathbb{R}^3$:

$\begin{bmatrix} -0.8 \\ 0.6 \\ \rule{1cm}{0.4pt} \end{bmatrix}, \begin{bmatrix} \rule{1cm}{0.4pt} \\ \rule{1cm}{0.4pt} \\ 1 \end{bmatrix}, \begin{bmatrix} \rule{1cm}{0.4pt} \\ 0.8 \\ \rule{1cm}{0.4pt} \end{bmatrix}$.

21. (1 pt) 2360lib/chap5/leon.5.3/ur_la.20.1.pg

Find the least-squares solution x^* of the system

$\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} x = \begin{bmatrix} -9 \\ -2 \\ -8 \end{bmatrix}$.

$x^* = \begin{bmatrix} \rule{1cm}{0.4pt} \\ \rule{1cm}{0.4pt} \end{bmatrix}$.

22. (1 pt) 2360lib/chap5/leon.5.3/ur_la.20.2.pg

Find the least-squares solution x^* of the system

$\begin{bmatrix} 1 & -2 \\ -1 & 2 \\ 5 & 4 \end{bmatrix} x = \begin{bmatrix} -12 \\ 10 \\ -13 \end{bmatrix}$.

$x^* = \begin{bmatrix} \rule{1cm}{0.4pt} \\ \rule{1cm}{0.4pt} \end{bmatrix}$.

23. (1 pt) 2360lib/chap5/leon.5.3/ur_la.20.4.pg

Find the least-squares solution x^* of the system

$\begin{bmatrix} 1 & -1 & -1 \\ 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{bmatrix} x = \begin{bmatrix} -12 \\ -6 \\ 2 \\ 0 \end{bmatrix}$.

$x^* = \begin{bmatrix} \rule{1cm}{0.4pt} \\ \rule{1cm}{0.4pt} \\ \rule{1cm}{0.4pt} \end{bmatrix}$.

24. (1 pt) 2360lib/chap5/leon.5.3/ur_la.20.5.pg

Fit a linear function of the form $f(t) = c_0 + c_1 t$ to the data points $(-6, -25)$, $(0, -1)$, $(6, 35)$, using least squares.

$c_0 = \rule{1cm}{0.4pt}$,
 $c_1 = \rule{1cm}{0.4pt}$.

25. (1 pt) 2360lib/chap5/leon.5.3/ur_la.20.6.pg

Fit a quadratic function of the form $f(t) = c_0 + c_1 t + c_2 t^2$ to the data points $(0, -8)$, $(1, -7)$, $(2, -18)$, $(3, -21)$, using least squares.

$c_0 = \rule{1cm}{0.4pt}$,
 $c_1 = \rule{1cm}{0.4pt}$,
 $c_2 = \rule{1cm}{0.4pt}$.

26. (1 pt) 2360lib/chap5/leon.5.5/n01.pg

(S. 5-5 cf. no. 1)

Which of the following sets of vectors form an orthonormal basis for $\mathbb{R}^2$?

(a) $\{(-1, 0)^T, (0, 1)^T\}$

- yes
- no

(b) $\{(-1, 1)^T, (1, -1)^T\}$

- yes
- no

(c) $\{(2/\sqrt{29}, 5/\sqrt{29})^T, (-5/\sqrt{29}, 2/\sqrt{29})^T\}$

- yes
- no

(d) $\{(4/\sqrt{7}, -3\sqrt{7})^T, (3/\sqrt{7}, 4/\sqrt{7})^T\}$

- yes
- no

(e) $\{(4/5, -3/5)^T, (3/5, 4/5)^T\}$

- yes

- no

27. (1 pt) 2360lib/chap5/leon_5.5/n03.pg
(S. 5-5 cf. no. 3)

$\left\{ \mathbf{u}_1 = \begin{pmatrix} 1/3 \\ 2/3 \\ -2/3 \end{pmatrix}, \quad \mathbf{u}_2 = \begin{pmatrix} 0 \\ 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix} \right\}$ is an orthonormal set of vectors in R^3 .. (You should verify this.)

Find the projection of the vector $\mathbf{v} = (1, 2, 3)^T$ onto the subspace S spanned by $\mathbf{u}_1$ and $\mathbf{u}_2$. (Write your answer in WeB-WorK's angular bracket notation for vectors.)
